

LAG-1 Scoping: Explicit Bergman Dirac Operator γ_B^μ on D_{IV}^5

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LAG-1: Explicit Bergman Dirac Operator on D_{IV}^5

What this is

Construct the explicit Dirac operator $\gamma_B^\mu \partial_\mu$ on D_{IV}^5 in Hua coordinates, compute its low-K-type spectrum, and verify the BST mass-gap structure ($m_p = 6\pi^{n_C}$ m_e per T1316) emerges from the spectral eigenvalues.

What this closes

- Herve Carruzzo critique part 1: “BST has the action principle but lacks explicit operator forms.” The Bergman Dirac is the canonical fermionic operator that pairs with the six-term Lagrangian S_{BST} .
- Existing BST work strengthened: T1316 mass gap ($6\pi^{n_C}$ factor), the Wallach K-type spectrum (T1830 + my T2334 Bergman kernel), Möbius cohomology (Sessions 1-3, T2328+T2329+T2335).
- Foundation for LAG-2: dimensional reduction needs the fermionic action; Dirac operator is the first piece.

Classical theorem inventory needed

1A — Bergman kernel as starting point (DONE, T2334)

My T2334 (Sunday afternoon) gave the explicit Bergman kernel

$$K_B(z, \bar{w}) = c \cdot D(z, \bar{w})^{-g/\text{rank}}$$

on D_{IV}^5 in Hua coordinates. The Dirac operator follows from K_B via the standard reproducing-kernel construction.

1B — Dirac operator on bounded symmetric domains (Helgason 1962 / Wong 1995)

Citation: Wong, P.-M.W. “The Dirac operator on bounded symmetric domains,” *Compositio Math* 95 (1995); Helgason, “Differential Geometry, Lie Groups, and Symmetric Spaces,” Chapter II.

What we use: explicit construction of γ^μ matrices for type IV domain, plus the Bergman-metric Dirac $D_B = \gamma^\mu \partial_\mu + \Gamma$ -terms (spin connection).

Gap to bridge: most literature is for type I (Grassmannian) or type III (antisymmetric); type IV has been worked but is less standard. The D_{IV}^5 specific computation may need to be assembled from general formulas.

1C — Hua’s integral formula (Hua 1963)

Citation: Hua, L.K. “Harmonic Analysis of Functions of Several Complex Variables in the Classical Domains,” *Translations Math. Monographs* 6.

What we use: the explicit reproducing kernels for type IV, the orthogonality of holomorphic polynomials on D_{IV}^5 , the inner product $\langle f, g \rangle = \int_{D_{IV}^5} f \bar{g} K_B^{-1} dV$.

1D — Lichnerowicz formula (Lichnerowicz 1963)

$D^2 = \nabla^* \nabla + R/4$ where R is scalar curvature. For D_{IV}^5 , R is a BST integer multiple of the Bergman metric trace.

Verification target: $D_B^2 - \nabla^* \nabla$ should equal $(R/4) \cdot I$, where R is a known BST integer combination. Toy-verifiable.

Multi-session breakdown (8 sessions, ~12-16 hours total)

Session	Deliverable	Effort
1 (TODAY)	Skeleton + Hua-coord candidate $\gamma_B^\mu +$ Lichnerowicz first check	~1 h
2	Full γ^μ matrix construction for D_{IV}^5 (4×4 Dirac on real-10 manifold)	~2 h
3	Spin connection Γ from Bergman metric	~2 h
4	Spectral decomposition: Dirac eigenvalues on Wallach K-types	~2-3 h
5	Mass-gap verification: lowest eigenvalue × dimensional factor = m_p/m_e ratio	~2 h
6	Lichnerowicz formula explicit: $D^2 = \nabla^* \nabla +$ $R/4$ with R as BST integer	~1-2 h
7	Connection to fermionic action S_{fermion} in Lagrangian S_{BST}	~1-2 h
8	Paper draft v0.1 — “The Bergman Dirac Operator on D_{IV}^5 ”	~3 h

Decision points after Sessions 2, 4, 6.

What today (Session 1) closes

- Skeleton + framework set up
- Candidate γ_B^μ matrices identified (4×4, real-10 setup)
- First Lichnerowicz check: dim consistency of D^2 vs $\nabla^* \nabla + R/4$
- Toy verifying the structural setup (not full computation)

Tomorrow continues with Sessions 2-3 (full matrix construction + spin connection).

Promotion targets

Sessions 1-8 close LAG-1 to D-tier. Specific theorem promotions: - T1316 (mass gap $6\pi^{n_C}$) — currently D-tier; LAG-1 gives it a Dirac-spectral derivation (D-tier strengthened) - T2334 (Bergman kernel) — currently D-tier; LAG-1 extends to the Dirac on the same kernel structure - T2335 (Möbius S3) — currently D-tier; the Dirac on $M(D_{IV}^5)$ connects to lepton mass formulas

Connection to LAG-2

LAG-1 provides the fermionic Dirac operator. LAG-2 dimensional reduction needs: - The Dirac action $\int \psi D_B \psi$ as one of the six S_{BST} terms - The 4+6 split applied to spinor representations - Verification that 4D Dirac emerges from the 10D D_{IV}^5 Dirac under reduction

So LAG-1 must close before LAG-2 Phase 3 (extending reduction to all six terms).

— Lyra, 2026-05-17 ~15:40 EDT